

Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

Achim BAUMGÄRTNER¹, Erik WEIGAND¹,
Gian Bruno TRUÖL^{1,2}, Mirko SONNTAG^{1,2}

¹University of Applied Sciences Esslingen, Esslingen University, Germany

²Fraunhofer Application Center Keim, Esslingen, Germany

acbait01@hs-esslingen.de, erwewi01@hs-esslingen.de,
gian.ledl-truol@hs-esslingen.de, mirko.sonntag@hs-esslingen.de

Submitted: 01 August 2026 | Revised: 25 March 2026 | Accepted: 01 August 2026

Abstract. This papers goal is the investigation on how bidirectional charging stations can be distributed across a dynamic area by combining traffic simulation with partially extracted grid data from official sources. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility, traffic flow, and waiting time. The proposed workflow integrates mobility traces, station utilization forecasts, and synthetic grid constraints to identify robust deployment strategies. Results indicate that jointly optimizing traffic and grid criteria can reduce peak grid load while improving charging availability in high-demand districts.

Keywords: Bidirectional charging, traffic simulation, SUMO, OpenStreetMap, spatio-temporal analysis, charging demand, V2G

Nomenclature

BEAM	Behavior, Energy, Autonomy, and Mobility
EV	Electric Vehicle
MATSim	Multi-Agent Transport Simulation
POI	Point of Interest
SoC	State of Charge
SUMO	Simulation of Urban Mobility
TraCI	Traffic Control Interface
V2G	Vehicle-to-Grid
OSM	OpenStreetMap
CS	Charging Station

1. Introduction

The rapid growth of electric mobility is shifting charging-infrastructure planning from static placement rules toward data-driven decision support [1]. Bidirectional charging adds a further dimension, since vehicles are no longer only energy consumers but can also feed power back to the grid, a concept known as Vehicle-to-Grid (V2G), which turns charging points into distributed flexibility assets [2]. Deciding where to place such stations therefore requires understanding where and when charging demand actually concentrates.

Because electric vehicles still represent a small share of the fleet, there is little real-world data on where charging stations should be placed and how intensively they are used. At the same time, uncoordinated charging concentrates electrical load in space and time and can stress local distribution grids [3], which makes the spatial

and temporal structure of demand a central planning question. Existing siting studies, however, often rely on data or tooling that is not openly reproducible [4]. We therefore ask: *how can candidate locations for charging stations be ranked from openly available data, based on their accessibility and on how charging demand is distributed over space and time?*

To address this, we build a simulation workflow on openly available data. Road-network and land-use information are taken from OpenStreetMap (OSM), and traffic is simulated microscopically in Simulation of Urban Mobility (SUMO), controlled through its Python interface Traffic Control Interface (TraCI). From land-use Point of Interests (POIs)—residential buildings, offices, and amenities—we derive travel demand, and we model charging as a *spatiotemporal* process, that is, demand that varies simultaneously by location and time of day. The model distinguishes public charging stations from private home wallboxes with vehicle-specific access control. A custom web interface, which is inspired by SUMO’s web interface, lets users select an area of interest on a map and generate ready-to-run scenarios directly, lowering the barrier to reproducing and extending the analysis. Candidate sites are then assessed in terms of accessibility, station utilization, and the resulting local charging load.

The scope of this study is the planning and evaluation stage. We focus on the spatial and temporal distribution of charging demand and load under realistic mobility patterns. A detailed electrical model of the distribution grid (power flow, feeder and transformer constraints) as well as battery degradation and market or pricing mechanisms are beyond its scope and left to future work.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which can cause queues at popular stations while other sites remain virtually unused. At the same time, cars that are parked for extended periods of time (in residential or commercial areas) can remain plugged in, creating opportunities for bidirectional energy

exchange. The spatial distribution of charging stations therefore affects both user experience and grid load. For example, placing stations in high-traffic areas may improve accessibility but could also lead to congestion and increased waiting times. Conversely, locating stations in less busy areas might reduce queues but could limit their utilization and the potential for vehicle-to-grid (V2G) services. Understanding the spatial and temporal patterns of charging demand is therefore a prerequisite for infrastructure planning. However, there is a lack of open, reproducible methods for assessing candidate locations based on real-world mobility patterns and grid constraints.

1.2 Problem Statement

Given the aforementioned challenges, the problem statement of this paper can be formulated as follows: How can we determine the optimal spatial distribution of bidirectional charging stations, while regarding existing charging stations, in an urban area by integrating traffic simulation data with partially synthetic grid constraints, in order to maximize accessibility and minimize waiting times for electric vehicle users?

1.3 Contributions

This paper contributes a practical and reproducible workflow for planning bidirectional charging infrastructure with coupled traffic and grid evaluation. This work contains a user interface for deterministic scenario-based traffic generation along with a thorough analysis of candidate station portfolios under varying demand and grid conditions. By integrating synthetic grid constraints with mobility-based accessibility metrics, we provide a holistic framework for siting decisions that can be adapted to different urban contexts and policy objectives.

The main contributions of this work are as follows:

- A simulation-based siting pipeline that jointly evaluates mobility accessibility and distribution-grid impact.

- A multi-criteria scoring framework for comparing candidate station portfolios under proto-realistic demand variability.
- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.

2. Related Work

Planning charging infrastructure has been studied extensively, and recent reviews organize the field along the dimensions of station siting, sizing, grid reinforcement, trip-considerations and the treatment of uncertainty [5, 13]. Most contributions optimize either where demand is best served or how the grid copes with the additional load. Bidirectional charging, in which vehicles also feed power back, is only beginning to be integrated into such planning frameworks [9].

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices, often placing stations so as to maximize coverage of traffic along routes. Data-driven analyses suggest that charging demand is largely shaped by proximity to amenities, Electric Vehicle (EV) density, and adjacency to high-traffic corridors, which motivates deriving demand from land-use data as we do [6]. Microscopic traffic simulators are increasingly used for this purpose. SUMO in particular provides built-in energy and charging-infrastructure models and has been used to assign electric vehicles to charging stations from simulated traffic data [7, 8].

A different approach than SUMO, is a mesoscopic traffic simulation model that uses a network of nodes and links to represent the road network, and simulates traffic flow based on traffic demand and road capacity. This approach can be used to estimate charging demand at a more aggregated level, and can be useful for planning charging infrastructure in areas with limited data availability or where detailed traffic simulation is not feasible. Multi-Agent

Transport Simulation (MATSim) is an example of a mesoscopic traffic simulation model that has been used for this purpose, and paired with Behavior, Energy, Autonomy, and Mobility (BEAM) it is possible to simulate the impact of charging infrastructure on a dynamic traffic network [11, 12].

2.2 Grid-Aware Siting

Grid-aware planning approaches prioritize feeder hosting capacity, voltage stability, and peak-load mitigation, frequently relying on metaheuristic optimization to jointly site and size stations under electrical constraints [5]. While they improve technical feasibility, they can yield a poor user experience when travel behavior is not modeled in sufficient detail. The placement of charging stations is often motivated not only by charging demand and by economic advantages, but also by repercussions for the electrical network [14]. While a reasonable amount of studies have been conducted on the siting of charging stations, there are only few that consider working with large-scale electric vehicle charging infrastructure [15].

2.3 Coupled and Bidirectional Approaches

A smaller but growing body of work couples transportation simulation with power-system analysis, for example through co-simulation frameworks that pass simulated charging loads to a distribution-grid model [10]. For bidirectional charging, planning models increasingly distinguish public and private chargers and account for uncertainty in demand and renewable generation [9]. Integrated and openly reproducible frameworks that combine microscopic mobility simulation with bidirectional charging at the level of individual public stations and private home wallboxes, however, remain limited. This paper addresses that gap on the mobility-and-demand side, while leaving a detailed electrical model of the distribution grid to future work.

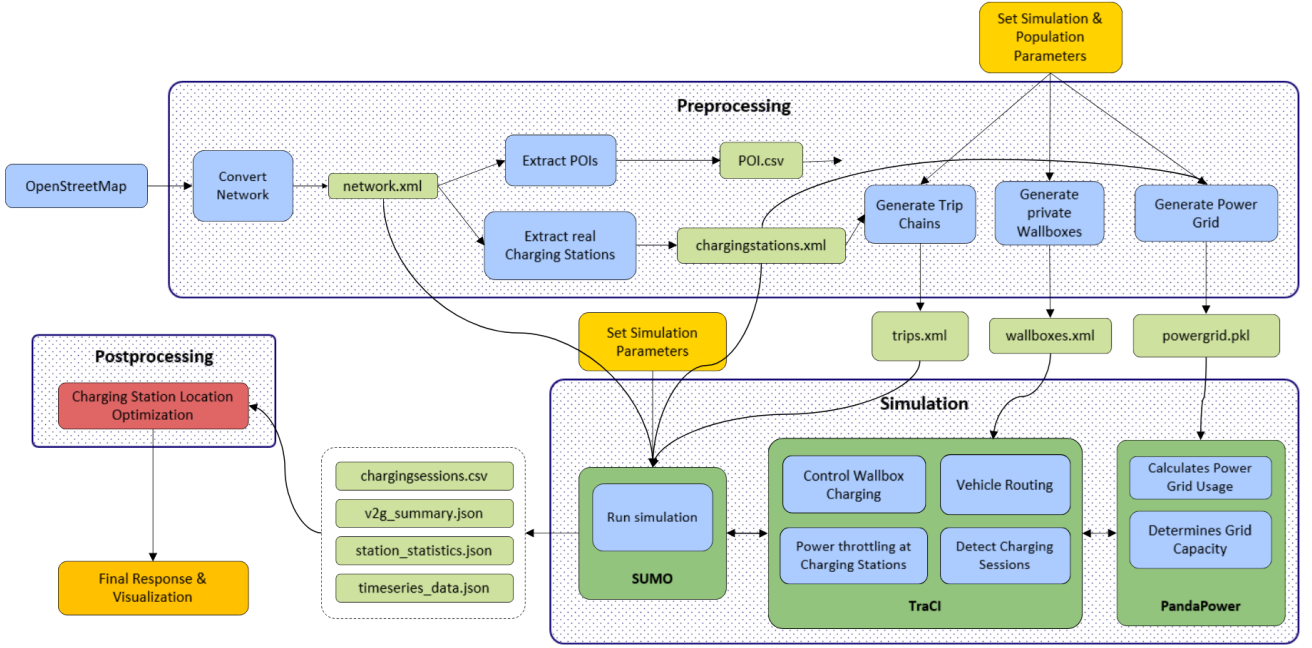


Figure 1: Complete Workflow Pipeline: From OSM data acquisition to scenario generation, execution and manipulation to analysis and visualization.

3. Methodology

SUMO is used as the microscopic traffic simulator and TraCI for real-time control, combined with a grid-impact model and a statistical distribution of home-charging devices, to evaluate the spatial distribution of bidirectional charging stations. Two siting strategies – *Saturate-and-Prune* and *Clustering* – are implemented and detailed in section 3.2.

3.1 System Overview

Figure 1 illustrates the complete workflow pipeline. The framework is organized into four consecutive stages: (i) data acquisition, (ii) preprocessing and scenario generation, (iii) coupled traffic-grid simulation, and (iv) analysis and visualization. Each stage produces intermediate artifacts that are consumed by the subsequent stage, so that the entire process is reproducible from a single set of user-defined parameters.

Data acquisition. The pipeline retrieves information like the road network, electrical grid topology, and candidate POIs from OSM. The road network is converted into a SUMO-

compatible format using the `netconvert` tool, while grid features and POIs are persisted for later preprocessing. This ensures that the simulation has accurate and relevant data to work with, allowing for realistic modeling of traffic flow and charging behavior in the region of interest.

Preprocessing and scenario generation.

The `chargingstations.xml` file contains both, existing and candidate stations produced by the placement algorithm, and is utilized not only for the generation of the synthetic power grid, but also in assistance with the POI file to generate trip chains. Trip chains map individual agents to a sequence of activities and destinations, from which their charging demand and low-State of Charge (SoC) behavior emerge. In parallel, the private-wallbox generator assigns home-charging devices to a share of agents following a statistical distribution, and the power-grid generator couples the charging stations and wallboxes to their nearest buses to form the synthetic distribution grid. To finalize the preprocessing, user-set parameters (EV penetration rate, wallbox share, battery capacities, grid voltage levels) are applied to the mecha-

nisms of the trip chain, private wallbox, and power grid generation, yielding a fully specified scenario. Scenario generation is seeded and therefore deterministic; the coupled simulation is deterministic only once the interpreter's hash seed is fixed as well, as discussed in section 4.

Coupled traffic-grid simulation. After the preprocessing the files `osm.passenger.trips.xml`, `private_wallboxes.xml`, and `power_grid.pkl` will then be used by SUMO to simulate traffic flow and charging behavior. TraCI, SUMO and PandaPower are the key components of the simulation, which is executed in a time-stepped manner. TraCI allows for real-time control of the simulation, enabling the dynamic adjustment of traffic flow and charging behavior based on an Application Programming Interface (API) via Python. SUMO simulates traffic on a microscopic level (i.e., individual vehicles and their interactions), while PandaPower evaluates the impact of charging behavior on the synthetic distribution grid. While SUMO and TraCI handle the traffic simulation, SUMO and PandaPower do not communicate directly. Instead, TraCI acts as a bridge between the two, allowing for the exchange of information and control signals. This enables the simulation to capture the complex interactions between traffic flow and grid behavior, providing the ability for SUMO to incorporate grid constraints and for PandaPower to calculate the impact of charging behavior on the grid.

Analysis and visualization. The simulation produces a variety of output files, including `station_statistics.json`, `v2g_summary.json`, `charts_data.json`, `charging_sessions.csv` and `model_log_data.csv`. These persisted outputs are analyzed and visualized to evaluate the placement algorithm along metrics such as charging sessions, net energy flow over time, peak grid power, and individual charging station utilization (section 3.3). The results are rendered as plots and maps in the web

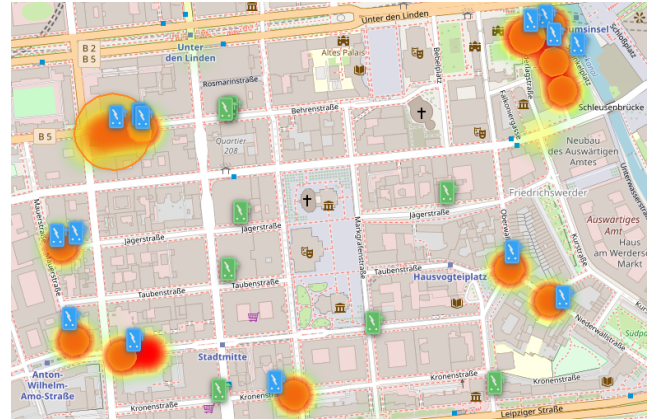


Figure 2: Resulting visualization with generated (blue) and extracted (green) CS for example run.

interface, and inform future iterations of the placement algorithm. Figure 2 shows an example visualization with generated (blue) and extracted (green) charging stations for a single run. The heatmap and orange circles indicate approximate areas for charging station placement, while the blue and green dots represent the actual locations of the generated and extracted charging stations, respectively.

3.2 Charging Station Placement Algorithm

Two siting strategies are implemented to compare different placement policies:

- **Saturate-and-Prune:** A deliberately naive iterative approach. Candidate stations are first placed on every road segment long enough to hold one, which guarantees maximum coverage but far exceeds any plausible build-out. The scenario is then simulated, and every candidate that carried less than 5% of the energy delivered by all public stations in that run is removed before the next iteration. Utilization is measured as a share of delivered energy rather than as occupied time, so that a station visited often but briefly is not mistaken for a well-used one. Stations that already exist in the area are exempt from pruning, because existing infrastructure cannot be removed in practice and the strategy is meant to decide where to

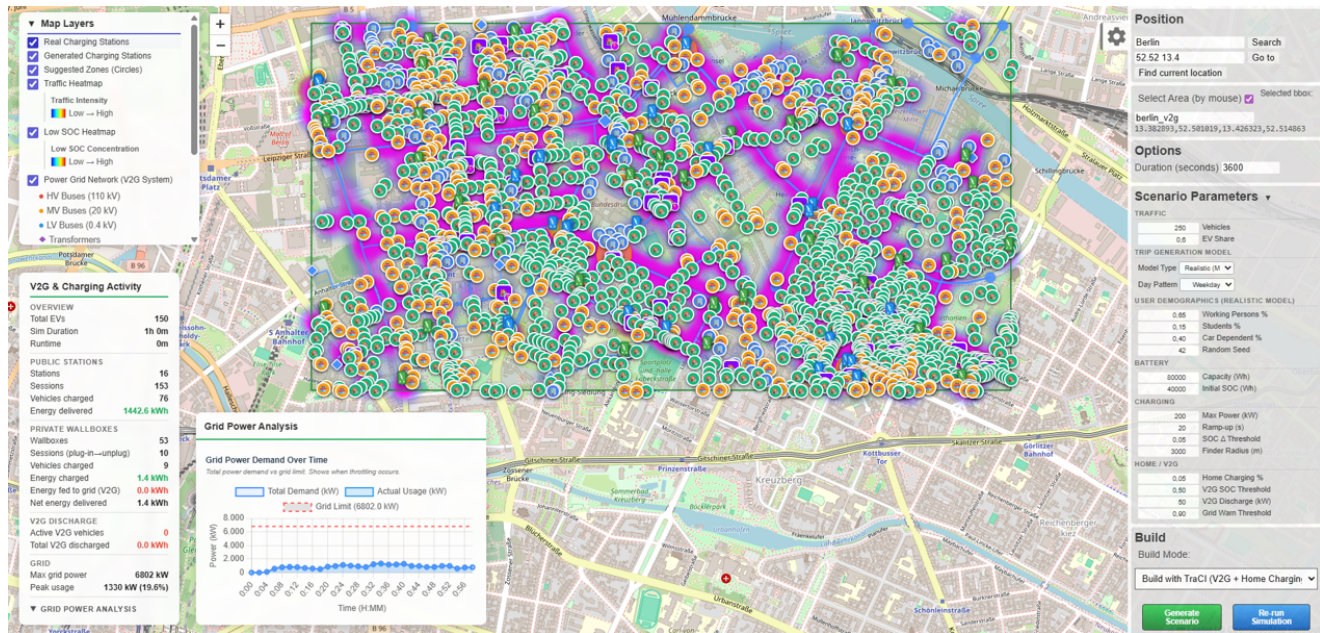


Figure 3: Example run of the workflow: the web interface with all data layers enabled.

add capacity. The loop is run for a fixed number of iterations rather than to convergence, since each iteration requires a full simulation.

- **Clustering:** A heuristic that prioritizes locations with high traffic flow, potential traffic congestion, and consideration for vehicles that enter a state of low SoC. Placement follows the traffic flow, aiming to maximize accessibility where vehicles are most likely to need charging while reducing detours.

3.3 Evaluation Metrics

Mobility accessibility. CS-placement is evaluated based on traffic flow, charging behavior, energy consumption, and SoC. The gathered data is used to calculate the accessibility of the charging stations, which is defined as the percentage of agents that are able to access a charging station within a certain distance or time threshold. Distance and time are key factors in determining the accessibility of charging stations, as they directly impact the convenience and feasibility of using the stations for charging. Distance is an honest measure of how far an agent must travel to reach a charging station, while time takes into account factors such as

traffic conditions, congestion, timely availability of charging stations, time and power (kWh) required to charge.

Grid impact. During the simulation, PandaPower solves an AC power flow on the synthetic distribution grid at every step, evaluating bus voltages and the loading of lines and transformers as constraints that bound the power allocated to each station. The grid impact reported per run is the aggregate demand and the peak grid power relative to the available grid capacity, which captures the stress the charging activity imposes on the grid without exceeding its voltage and loading limits.

Aggregation. The evaluation metrics are aggregated to provide a comprehensive assessment of the charging station placement algorithm. This includes combining mobility accessibility and grid impact metrics to evaluate the overall performance of the charging station network. The aggregation process involves analyzing the trade-offs between accessibility and grid impact, allowing for a balanced evaluation of the placement algorithm's effectiveness.

4. Experiments

4.1 Datasets

The experiment is conducted on mobility data, which is generated by an algorithm that simulates movement patterns of agents in a given area. The algorithm takes into account various factors such as land use, transportation infrastructure and natural behavior of agents to generate realistic mobility data. The generated data is then used to evaluate the performance of the proposed charging station placement algorithm. Before running the simulation, the dataset consists of surface level mobility (agent start- and endpoints, OSM-data), which is used for traffic flow, POIs and electrical grid information. When a simulation is run, the dataset is enriched with additional information such as charging behavior, energy consumption, specific locations of agents and their SoC. The datasets are generated using a combination of real-world data and synthetic data, ensuring that the results are both realistic and generalizable to other urban areas. The datasets are also designed to be scalable, allowing for the evaluation of the placement algorithms on larger areas with more agents and charging stations.

4.2 Experimental Setup

The reference scenario covers a one kilometer by one kilometer area in Berlin, Germany. Each agent follows a daily trip chain with departures spread across the day and returns to its home location between activities. The simulation covers 24 hours of simulated time with a time step of 10 s, so the chains complete within the horizon and the evening period at home is part of the observation. The trip generator is selected by the pipeline from the configured duration: horizons below ten hours use a compressed departure window intended for short test runs, longer horizons use the daily generator applied here.

Every experiment is repeated over five random seeds, which redraw home and activity locations, the assignment of electric vehicles, and departure times. Across those five popula-

tions the scenario holds 236.6 ± 3.4 agents, of which 142.6 ± 1.8 are electric vehicles with a battery capacity of 80 kWh initialised at 50 % SoC; the remainder are conventional vehicles. Every agent carries a SUMO battery device, because the simulation is configured with `device.battery.probability = 1`, so the fleet-wide SoC statistics cover all agents that entered the network, while charging is restricted to the electric vehicles. Half of the electric vehicle owners are assigned a private home wallbox at their home lane. The wallboxes are deliberately excluded from the additional file handed to SUMO and exist only inside TraCI, which restricts each of them to its owning vehicle type. As a result SUMO's stationfinder cannot route any vehicle to a private wallbox, and home charging occurs only when an agent is parked at its own home location.

Charging power, V2G discharge and the wallbox logic are injected at runtime through TraCI, which overrides the station power, applies the ramp-up, and arbitrates the total power draw against a synthetic distribution grid modelled in PandaPower with 37 buses, 93 lines and 10 transformers. Charging power per station is capped at 200 kW with a 20 s ramp-up, V2G discharge is enabled above 50 % SoC with up to 50 kW per vehicle and is triggered when the grid reaches 90 % of its capacity. The grid capacity is derived from the area of the scenario at a density of 1500 kW per square kilometer, giving 1500 kW here.

The public infrastructure of the reference case is the one already present in the area, that is, the 26 charging stations extracted from OSM. Each station is projected onto the nearest road segment whose lane is long enough to hold it, which keeps all 26 stations in the network: placing them on the closest segment regardless of its length puts several of them on junction stubs of a few decimetres, where the resulting extent is degenerate and rejected.

Reproducibility. Repeated executions of an identical configuration were initially found to differ, with a spread of roughly 7 % in public and 9 % in private charging energy. The cause is

the iteration order over the set of active electric vehicles, which determines the order in which vehicles request power from the shared pool and which varies between processes because Python randomises string hashing. Fixing the interpreter hash seed makes a run bit-identical, so all runs reported here are executed with a fixed hash seed and the variation between them is due to the scenario seed alone.

4.3 Results

4.3.1 The existing infrastructure

Where the energy is drawn. Over the simulated day the fleet draws 3853 kWh, of which public stations deliver 3251.7 ± 315.4 kWh in 246.2 ± 9.2 sessions to 98.2 ± 1.5 vehicles and private wallboxes deliver 601.5 ± 258.2 kWh in 27.0 ± 6.5 sessions. Public charging thus accounts for 84 % of the energy. The imbalance follows from how the two kinds of infrastructure are reachable. Public stations are visible to the stationfinder and are approached whenever a vehicle runs low on charge, whereas a wallbox can only be used when its owner happens to be parked at home, which the spread of departure and return times makes comparatively rare within the horizon. The large relative spread of the wallbox figures, 43 % of their own mean, reflects exactly that: whether home charging happens at all depends on which agents a given seed places where. Every wallbox session is attributable to the owning agent, so the type restriction on private stations holds.

Accessibility. Despite the public charging that does take place, the fleet ends the day well below its starting point. The mean SoC drops from 50.0 % to 28.0 ± 0.7 %, with a median of 33.9 ± 2.9 %. Of all agents, 43.9 ± 1.2 % end below 20 % SoC, 39.0 ± 2.2 % drop below 10 % at some point during the day, and 3.8 ± 0.3 % are fully depleted. Table 1 summarises the reference case.

Bidirectional operation. V2G discharge is enabled but does not occur in the reference case. The grid reaches a peak draw of $1085.5 \pm$

Table 1: Reference case on the existing OSM infrastructure, mean $\pm$ standard deviation over five scenario seeds. 24 hours of simulated time, 10 s time step, coupled through TraCI.

Metric	Value
Agents	236.6 ± 3.4
of which EV	142.6 ± 1.8
Simulation horizon	24 h (8640 steps)
Mean SoC, start $\rightarrow$ end	$50.0 \rightarrow 28.0 \pm 0.7$ %
Median end SoC	33.9 ± 2.9 %
Agents below 20 % SoC at end	43.9 ± 1.2 %
Agents below 10 % SoC (any time)	39.0 ± 2.2 %
Agents fully depleted	3.8 ± 0.3 %
Public CS available (OSM)	26
Public sessions	246.2 ± 9.2
Public vehicles served	98.2 ± 1.5
Public energy	3251.7 ± 315.4 kWh
Wallbox sessions	27.0 ± 6.5
Wallbox energy	601.5 ± 258.2 kWh
V2G energy discharged	0.0 kWh
Peak grid draw	1085.5 ± 149.1 kW

149.1 kW against a capacity of 1500 kW, so the discharge condition of 90 % utilisation is never met. The scenario does not put enough load on the grid for bidirectional operation to be called upon: roughly 143 electric vehicles distributed over one square kilometer, charging at different times of day, never coincide closely enough to stress a grid dimensioned for the area. The run therefore characterises charging demand and demonstrates that the mechanism is in place, but it cannot quantify a V2G benefit. This changes once stations are added, as section 4.3.3 shows.

4.3.2 Derived charging demand

The DBSCAN stage of the Clustering strategy operates on the samples in which a vehicle is below 30 % SoC; half of them are drawn at random to bound the clustering cost. Clusters that fall within 100 m of an already existing station are discarded, and across the five seeds the stage retains 34.6 ± 5.0 candidate sites. A sample is one vehicle at one 10 s time step, so the sample count measures dwell time rather than demand, and a vehicle parked at home for hours contributes thousands of samples without ever asking for

Table 2: The ten charging-demand clusters with the highest energy demand for one representative seed. *Vis.* is the number of distinct vehicle visits, *Veh.* the number of vehicles behind them, *kWh* the energy those visits need to reach 80 % SoC, *CP* the resulting number of charge points and *Cap.* the grid capacity at the nearest connection point.

ID	Vis. (#)	Veh. (#)	SoC (%)	Rad. (m)	kWh	CP (#)	Cap. (kW)
44	17	14	17.8	47.2	845.5	2	500
35	10	10	17.9	25.0	503.6	1	5000
13	9	9	15.2	32.8	468.8	1	5000
34	7	7	24.9	25.0	352.7	1	5000
45	7	7	22.5	65.3	351.8	1	5000
14	7	7	18.1	45.8	348.5	1	5000
37	5	5	22.6	25.0	250.1	1	5000
53	4	4	17.6	25.0	198.8	1	5000
24	4	4	16.9	48.2	196.7	1	5000
39	4	4	19.0	25.0	193.6	1	5000

All 32 clusters: 111 visits, 5657.4 kWh, 33 charge points; SoC $15.6 \pm 6.0\%$; radius 28.7 ± 9.4 m; grid rating 15 excellent, 17 good.

a public charge. The demand of a cluster is therefore derived from the vehicles themselves. Samples inside a cluster are grouped per vehicle and split into separate visits whenever a vehicle is absent for more than five minutes, and each visit is charged with the energy it would need to reach 80 % SoC. Table 2 lists the ten strongest clusters of one representative seed. Demand is spread over many sites rather than concentrated in a few, and the mean SoC inside a cluster is well below the 30 % threshold, which indicates that the retained clusters mark genuine range emergencies rather than opportunistic stops. Cluster radii stay compact, so each cluster is small enough to be served by a single station site.

4.3.3 Comparison of the placement strategies

Saturate-and-Prune places a candidate on every road segment long enough to hold one, which yields 196 candidates on top of the 26 existing stations, and then removes the least utilised half of the remaining candidates in each it-

eration. Running the loop for six iterations traces the network down from 222 to 33 stations and yields an accessibility curve against station count rather than a single operating point. Clustering derives its candidates from the demand log of the reference run in a single pass and ends at 60.6 ± 5.0 stations, which places it between the fourth and third iteration of that curve. Table 3 reports both.

Adding stations closes most of the gap, and the method used to add them does not matter. Any configuration with more than about fifty stations raises the mean end SoC from 28.0 % to roughly 31 %, lifts the median by ten points, and reduces the share of agents ending below 20 % SoC from 43.9 % to between 35 and 37 %. These effects are three to eight times the spread across seeds and are the clearest result of this study. At its own station count, however, Clustering lands squarely on the Saturate-and-Prune curve. Interpolating that curve to 60.6 stations gives a mean SoC of roughly 30.9 % against the $31.4 \pm 1.0\%$ Clustering achieves, a difference of half a standard deviation, and on the two tail metrics the ordering reverses: Clustering leaves $37.4 \pm 1.7\%$ of agents below 20 % and $31.7 \pm 2.4\%$ below 10 % where the interpolated curve gives roughly 36.6 and 29.9 %. Within the resolution of five seeds, the two strategies are equivalent, and what governs accessibility is how many stations are added rather than where a heuristic decides to put them.

Accessibility saturates well below full coverage. From 222 stations down to 75 the mean end SoC does not move at all, staying between 31.1 and 31.3 %, and the median even improves slightly as stations are removed. Degradation begins only below roughly fifty stations: at 39 stations the mean falls to $29.8 \pm 0.9\%$ and the share of agents dropping below 10 % SoC rises from 28.9 to 33.3 %. The useful operating range for this scenario is therefore between about fifty and seventy-five stations. Everything above it is redundant, which is why the fully saturated network of 222 stations performs no better than

Table 3: Accessibility against station count over five scenario seeds, mean $\pm$ standard deviation. The Saturate-and-Prune rows are successive prune iterations of the same loop; Clustering reaches its network in a single pass. Interpolating the Saturate-and-Prune curve to 60.6 stations gives the operating point at which the two strategies are directly comparable.

Configuration	Stations	Sessions	Energy (kWh)	Mean SoC	Median SoC	<20 %	<10 %	Pe
Existing infrastructure	26	246.2 $\pm$ 9.2	3252 $\pm$ 315	28.0 $\pm$ 0.7	33.9 $\pm$ 2.9	43.9 $\pm$ 1.2	39.0 $\pm$ 2.2	108
Saturate-and-Prune, iter. 1	222	315.6 $\pm$ 19.0	4213 $\pm$ 356	31.1 $\pm$ 1.2	42.7 $\pm$ 3.2	36.3 $\pm$ 3.4	28.1 $\pm$ 3.6	180
Saturate-and-Prune, iter. 2	124	312.6 $\pm$ 16.5	4264 $\pm$ 352	31.3 $\pm$ 1.1	43.9 $\pm$ 2.9	35.6 $\pm$ 3.3	28.2 $\pm$ 3.9	181
Saturate-and-Prune, iter. 3	75	304.0 $\pm$ 18.9	4237 $\pm$ 458	31.2 $\pm$ 1.1	44.0 $\pm$ 2.4	36.1 $\pm$ 2.5	28.9 $\pm$ 4.0	175
Saturate-and-Prune, iter. 4	51	302.6 $\pm$ 26.1	4376 $\pm$ 551	30.6 $\pm$ 1.1	43.8 $\pm$ 3.9	37.1 $\pm$ 2.6	30.5 $\pm$ 3.2	164
Saturate-and-Prune, iter. 5	39	290.6 $\pm$ 24.4	4304 $\pm$ 323	29.8 $\pm$ 0.9	39.1 $\pm$ 5.0	39.4 $\pm$ 2.0	33.3 $\pm$ 2.5	156
Saturate-and-Prune, iter. 6	33	276.2 $\pm$ 17.3	3971 $\pm$ 226	29.6 $\pm$ 1.1	40.6 $\pm$ 6.2	39.4 $\pm$ 2.3	33.2 $\pm$ 3.7	134
Clustering	60.6 $\pm$ 5.0	297.8 $\pm$ 18.6	5781 $\pm$ 530	31.4 $\pm$ 1.0	46.3 $\pm$ 1.7	37.4 $\pm$ 1.7	31.7 $\pm$ 2.4	200

the 75 that survive three prune steps, and it is also why both strategies converge on comparable outcomes: they both land inside that plateau.

Clustering is the only configuration that reaches the regime bidirectional charging is meant for. At comparable accessibility Clustering moves 5781 ± 530 kWh through its public stations against roughly 4300 kWh anywhere on the Saturate-and-Prune curve, and it drives the peak grid draw to 2006 ± 311 kW against 1640 to 1811 kW. Placing stations where vehicles are already at low SoC produces longer and more energetic sessions than distributing them along the road network. The resulting load is not a defect of the strategy but the condition under which flexibility becomes worth having: against a grid dimensioned for 1500 kW, Clustering exceeds the nominal capacity in all five seeds, at 114 to 158 % of it, whereas the reference case never even reaches the 90 % mark at which discharge is enabled. Solving the siting problem is what creates a grid situation that V2G could answer.

The binding constraint on V2G is fleet SoC, not grid load. Although Clustering brings the grid above its capacity in every seed, the discharged energy stays negligible: two of five seeds trigger discharge at all, for at most 0.8 kWh. The reason is visible in the accessibility figures. Discharge requires a vehicle above

50 % SoC, while the fleet ends the day at a mean of 31.4 % and the peak load occurs precisely when many vehicles are drawing power because they are low. Supply of flexibility and demand for it are anti-correlated in this scenario, so the stations that could feed back are occupied by vehicles that need to charge. This is a more specific finding than the reference case allowed: raising EV penetration alone would not unlock V2G, because it raises both sides at once. What the mechanism needs is a fleet segment that is parked with a high state of charge during the peak, which in practice means the private wall-boxes at home rather than the public stations on the road.

What separates the strategies is cost of computation, not quality of placement. Clustering obtains its network from one simulation of the reference case. Saturate-and-Prune needs one simulation per prune iteration, six in the configuration reported here, and the station count it arrives at is determined by how many iterations are run rather than by its 5 % utilisation criterion, which almost never binds. For a planning workflow that difference is substantial. It is, on this evidence, the only defensible advantage of the demand-driven heuristic over naive saturation.

The utilisation threshold does not govern the outcome. In the saturated network only one or two of 222 stations exceed 5 % oc-

cupancy, so the prune criterion selects almost everything for removal and the cap of one half per iteration is what actually determines the schedule. A sweep over the resulting networks shows why a different number would not repair this: 4.8 ± 2.6 stations lie above 5 % occupancy, 43.6 ± 1.5 above 1 %, and 48.6 ± 0.5 above 0.1 %. The distribution is bimodal, a station is either used or never touched, so any threshold between 1 % and 0.1 % yields nearly the same network. Saturate-and-Prune as implemented is thus better described as repeated halving guided by utilisation ranking than as a threshold rule.

5. Discussion

5.1 Practical Example

5.2 Limitations

Station placement on short lanes. Charging stations extracted from OSM are point locations that have to be projected onto a lane of the road network. Projecting each station onto the closest segment regardless of its length places several of them on junction stubs of a few decimetres, where the station extent collapses; SUMO rejects such a station and, because the rejection aborts parsing of the additional file, silently discards every station declared after it. The generator therefore selects the nearest segment that can hold the full station footprint. This keeps all stations in the network at the price of a small positional offset, which is acceptable at the resolution of the siting question but would matter in a study of access from individual parking spaces.

Bidirectional operation is barely exercised. V2G discharge is implemented and enabled, but contributes essentially no energy. In the reference case the peak draw of 1085 kW against a capacity of 1500 kW never approaches the 90 % discharge condition. Under Clustering the grid does exceed its nominal capacity in every seed, yet discharge still triggers in only two of five and delivers at most 0.8 kWh. The obstacle is therefore not the grid load but the state of the fleet: discharge requires a vehicle

above 50 % SoC, and the peak coincides with a fleet whose mean SoC is around 31 % because the vehicles at the stations are there to charge. Raising EV penetration would lift both sides of that balance and need not help. Quantifying a V2G benefit requires a configuration in which a distinguishable segment of the fleet is parked at a high state of charge while the peak occurs, which points at home charging during the evening rather than at public stations, and is left to future work.

Reachability of private wallboxes. Private wallboxes are restricted to their owning vehicle and are deliberately hidden from SUMO, so that no foreign vehicle can occupy them; the run confirms that the restriction holds, with all 20 wallbox sessions attributable to the owning agent. The same design also means that SUMO's stationfinder never routes a vehicle home to charge. Home charging occurs only when an agent is parked at its own home location for other reasons, which the spread of departure and return times makes rare within a 24-hour horizon. The low private share reported in section 4.3 therefore reflects the opportunity for home charging within this scenario, not a general statement about the role of private infrastructure.

Resolution of the comparison. All results are averaged over five scenario seeds, which is enough to separate both placement strategies from the reference case but not enough to separate them from each other: their accessibility differences are smaller than the spread across seeds. Reporting one as superior would require either more seeds or a scenario in which the two produce more distinct networks. The station count, on which they differ by 19 %, is the one dimension on which the present data does support a ranking.

One area, one fleet size. Every figure comes from the same square kilometer in Berlin with roughly 237 agents. The seeds redraw where those agents live and travel, not how many there are or what the road network looks like,

so the reported spread quantifies population variation within one scenario and says nothing about transfer to other districts or fleet sizes. The claim that the datasets are generalizable to other urban areas remains untested.

Charge-point sizing. The heuristic sizes a cluster by dividing its energy demand by the length of the observation window. Over a 24-hour horizon this averages away the peaks at which charging demand coincides, and consequently sizes every cluster at a single charge point regardless of how much energy it carries. A peak-based rule, applied to the busiest hour rather than to the daily mean, would give a more useful estimate.

Grid rating of candidate sites. Two different grid models are used in this work, and only one of them solves a power flow. During the simulation the grid controller runs an AC power flow at every step and reduces the power granted to a station when bus voltages leave the 0.95 to 1.05 pu band or a line or transformer exceeds its rating, so the reported load and peak respect the electrical limits of the synthetic grid. The grid rating attached to a *candidate* site in table 2, in contrast, is a static estimate: it takes the nominal capacity of the voltage level of the nearest bus, subtracts the load already connected to it, and grades the site by its distance to that bus. It therefore indicates proximity to a connection point of sufficient nominal capacity, not hosting capacity confirmed by a power flow along the feeder. A candidate rated *excellent* may still require reinforcement once the stations are actually operated.

6. Conclusion

This paper presented a simulation-driven method for distributing bidirectional charging stations under joint traffic and grid criteria. The evaluation demonstrates that coordinated planning can improve both charging accessibility and electrical reliability.

6.1 Future Work

Future work will include stochastic optimization under forecast uncertainty and integration of dynamic pricing signals. We also plan to validate the framework with larger real-world datasets and pilot deployments in mixed urban-suburban regions.

Acknowledgments

The authors thank the Fraunhofer Center of Application KEIM for supporting this research work.

References

- [1] International Energy Agency (IEA), *Global EV Outlook 2023: Catching up with Climate Ambitions*. Paris, France: IEA, 2023. [Online]. Available: <https://www.iea.org/reports/global-ev-outlook-2023>
- [2] W. Kempton and J. Tomić, “Vehicle-to-grid power fundamentals: Calculating capacity and net revenue,” *Journal of Power Sources*, vol. 144, no. 1, pp. 268–279, 2005.
- [3] K. Clement-Nyns, E. Haesen, and J. Driesen, “The impact of charging plug-in hybrid electric vehicles on a residential distribution grid,” *IEEE Transactions on Power Systems*, vol. 25, no. 1, pp. 371–380, 2010.
- [4] F. He, D. Wu, Y. Yin, and Y. Guan, “Optimal deployment of public charging stations for plug-in hybrid electric vehicles,” *Transportation Research Part B: Methodological*, vol. 47, pp. 87–101, 2013.
- [5] “A review on electric vehicle charging station planning: Infrastructure placement, sizing, upgrades, and uncertainties,” *Journal of Energy Storage*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352152X25040381>

- [6] “Data-driven optimization of EV charging station placement using causal discovery,” *arXiv preprint arXiv:2503.17055*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.17055>
- [7] T. Kurczveil, P. Á. López, and E. Schnieder, “Implementation of an energy model and a charging infrastructure in SUMO,” in *Simulation of Urban Mobility (SUMO)*, https://www.researchgate.net/publication/276063158_Implementation_of_an_Energy_Model_and_a_Charging_Infrastructure_in_SUMO,
- [8] “Assignment approach for electric vehicle charging using traffic data collected by SUMO,” *World Electric Vehicle Journal*, vol. 14, no. 2, art. 40, 2023. [Online]. Available: <https://www.mdpi.com/2032-6653/14/2/40>
- [9] “Vehicle-to-grid enabled charging infrastructure planning and operations considering demand uncertainties,” *Transportation Research Part D: Transport and Environment*, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1361920923003152>
- [10] “Assessing EV charging impacts on power distribution systems: A unified co-simulation framework,” *arXiv preprint arXiv:2505.21773*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.21773>
- [11] A. Horni, K. Nagel, and K. W. Axhausen, Eds., *The Multi-Agent Transport Simulation MATSim*. London, U.K.: Ubiquity Press, 2016. [Online]. Available: <https://doi.org/10.5334/baw>
- [12] C. Sheppard, R. Waraich, A. Campbell, A. Gopal, and A. Pozdnukhov, “Modeling plug-in electric vehicle charging demand with BEAM: The framework for behavior, energy, autonomy, and mobility,” Lawrence Berkeley National Laboratory, Berkeley, CA, USA, Tech. Rep. LBNL-2001018, 2017. [Online]. Available: <https://www.osti.gov/biblio/1398472>
- [13] R. Pagany, L. Ramirez Camargo, and W. Dorner, “A review of spatial localization methodologies for the electric vehicle charging infrastructure,” *International Journal of Sustainable Transportation*, vol. 13, no. 6, pp. 433–449, 2019. [Online]. Available: <https://doi.org/10.1080/15568318.2018.1481243>
- [14] F. Ahmad, A. Iqbal, I. Ashraf, M. Marzband, and I. Khan, “Optimal location of electric vehicle charging station and its impact on distribution network: A review,” *Energy Reports*, vol. 8, pp. 2314–2333, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352484722001809>
- [15] “Large-scale optimization of electric vehicle charging infrastructure,” in *Proc. 32nd ACM Int. Conf. on Advances in Geographic Information Systems (SIGSPATIAL ’24)*, 2024. [Online]. Available: <https://dl.acm.org/doi/10.1145/3678717.3700830>
- [16] V. Lazari and A. Chassiakos, “Multi-objective optimization of electric vehicle charging station deployment using genetic algorithms,” *Applied Sciences*, vol. 13, no. 8, art. 4867, 2023. [Online]. Available: <https://www.mdpi.com/2076-3417/13/8/4867>